THE UNIVERSITY *of* EDINBURGH

Bioinformatics Programming and System Management (PGBI11095)

ICA 1: MyFirstPipeline

Exam Number: B300238

GitHub Links

GitHub Repository Link: https://github.com/B300238-2025/MyFirstPipeline

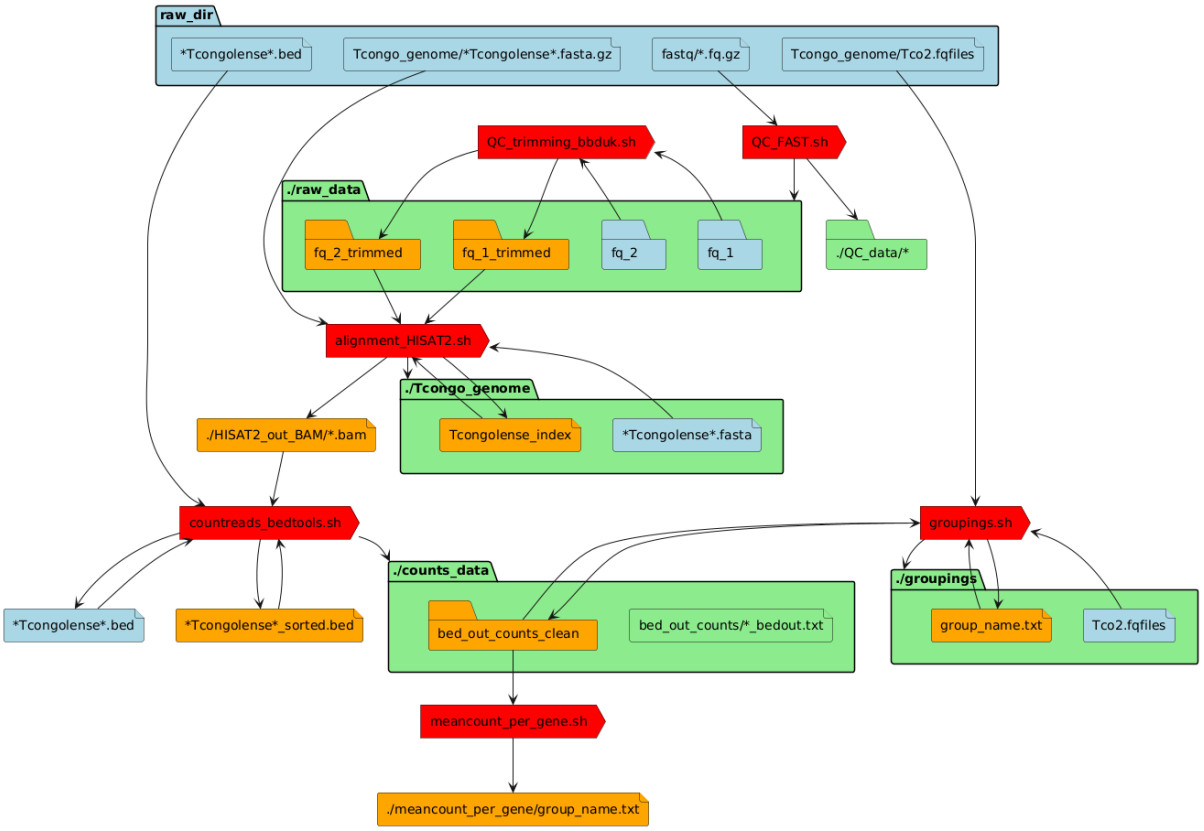
B3002382025.MyFirstPipeline.tar.gz.cpt Encryption Key: **Pipelin555**

How to run MyFirstPipeline?

1) Change working directory to dir containing all the .sh scripts
2) In terminal, type "./myfirstpipeline.sh"
3) Let it run!

Description

MyFirstPipeline contains 7 Shell scripts:

- myfirstpipeline.sh ➜ "Main" file that runs the rest of the scripts sequentially
- QC_FAST.sh
    - Runs FASTQC on raw .fq.gz files, extracting and deleting any zipped file
    - Copies raw data files into ./raw_data/fq_1 and ./raw_data/fq_2 dirs, and unzip
- QC_trimming_bbduk.sh ➜ Using bbduk sequence trimming script, trims paired-end sequences at both right and left sides, and trim bases with less than 10 Phred quality score
    - Why 10 Phred quality score? It is the standard benchmark for maximising alignment rates
- alignment_HISAT2.sh
    - Copies Tcongo genome into current working dir, and unzip to get .fasta file
    - Before alignment, builds Tcongolense_index with HISAT2 using *Tcongolense*.fasta file
    - Performs sequence alignment using HISAT2 for trimmed paired-end sequences using reference Tcongo genome, piping output to samtools for formatting .bam files
- countreads_bedtools.sh
    - Copies Tcongolense bedfile to current working dir
    - Before counting with bedtools, make sure to sort Tcongolense bedfile by gene name and 0-base start position (to lighten the workload for bedtools)
    - Uses bedtools coverage to count reads overlapping each gene in Tcongolense genome.
    - Uses awk to print useful cols (gene, geneid, description, counts)
- groupings.sh ➜ Using the reference raw file Tco2.fqfiles, group the clean counts data set by SampleType (Clone1, Clone2, WT), Time (0, 24, 48), and Treatment (Induced, Uninduced)
- meancount_per_gene.sh ➜ Iterates through each dir in bed_out_counts_clean, uses awk to merge the samples and calculate the mean count for each gene for each group (3/4 replicates)

For more details information on the inputs/outputs, please refer to comments in the shell scripts and the following flowchart: